Chapter 11 — Applications of self-supervised learning

Pretext tasks become concrete in domain settings

Learning objectives

- By the end of this part

Computer vision applications

- Segmentation before scarce labels are applied
- Which matters in domains where bounding boxes or masks are expensive

Data efficiency in vision

- Once representations exist
- Teams still validate on held-out gold labels

Natural language processing applications

- Masked language modeling and related objectives pretrain encoders on unlabeled text
- Downstream tasks such as classification, named entity recognition
- Large web-scale corpora supply the unlabeled fuel for that pretraining stage

Transfer to specialized text

- Domain teams often start from a general pretrained language model and fine-tune on a
- The pretext-learned context and syntax transfer
- This pattern is common when expert text labels are scarce

Time-series forecasting and imputation

- For time series
- Models learn temporal structure from unlabeled sensor or transaction streams
- Gap filling also prepares incomplete series for downstream monitoring tasks

Anomaly detection and contrastive sequences

- Useful for fraud or equipment failure
- Clustering without labeling every window

Takeaways

- Self-supervised applications span vision, language, and time series with a shared theme
- Benefits include data efficiency, real-time monitoring potential
- Quality still depends on validating the downstream labeled evaluation set

Next

- Complete the quiz for this part
- The next part compares crowdsourcing and expert annotation on scale, cost, turnaround